

Google Merchandise Store Sales Analysis

Business Analysis Report

1. Executive Summary

This project answers business questions about revenue, traffic, and customer behavior using Google Analytics records of Google Merchandise Store Sample data. The objective is to understand revenue performance, customer acquisition channels, product performance, and funnel conversion behaviour.

The analysis was performed using SQL in Google BigQuery and visualised using Looker Studio dashboards (formerly Data Studio).

Metric	Value
Total Revenue	\$964,592.82
Total Purchases	21,660
Product Views	362,607
Add to Cart Actions	121,427
Funnel Purchases	24,532
Purchase Conversion Rate	6.77%

2. Business Objective

The project answers key ecommerce questions to drive strategic decisions:

- **Revenue Trend:** How is revenue performing over time?
- **Traffic Channel:** Which traffic channels generate valuable customers?
- **Top Products:** Which products contribute the most revenue?
- **Geographic Revenue:** Which region generates the most revenue?

- **Funnel and Conversion Rate Analysis:** Where are customers dropping off in the purchase journey?

3. Dataset

Source: Google Analytics Sample Dataset

BigQuery Table: bigquery-public-data.google_analytics_sample.ga_sessions_*

The data is typical of what an ecommerce website would see and includes the following information:

- **Traffic source data:** Information about where website visitors originate, including data about organic traffic, paid search traffic, and display traffic.
- **Content data:** Information about the behavior of users on the site, such as URLs of pages that visitors look at, how they interact with content, etc.
- **Transactional data:** Information about the transactions on the Google Merchandise Store website.

Limitations: Data for some fields is obfuscated such as fullVisitorId, or removed such as clientId, adWordsClickInfo and geoNetwork. “Not available in demo dataset” will be returned for STRING values and “null” will be returned for INTEGER values when querying the fields containing no data.

During validation, it was found that product-level funnel metrics were inconsistent because of how ecommerce actions were stored in the Google Analytics sample schema. The analysis was then adjusted to an event-level funnel, which produced reliable business KPIs.

4. Tools and Technologies

- **SQL** for robust data querying and transformation.
- **Google BigQuery** as the cloud data warehouse solution and created BigQuery **Views** to connect to Looker Studio for reporting-ready datasets.
- **Looker Studio** for interactive reporting and visualization dashboards.

5. SQL Data Exploration and Analysis

The first stage of the project involved exploring the raw Google Analytics dataset and identifying relevant fields required for ecommerce analysis.

5.1 Revenue Trend & Performance Over Time

To understand overall website activity and macro growth patterns, historical website sessions and raw revenue metrics were explored across consistent time periods.

5.2 Traffic Channel Performance & Customer Value

Traffic performance was analysed by extracting traffic medium information to isolate high-value user pathways. Calculated metrics included:

- Sessions by traffic channel
- Transactions by traffic channel
- Conversion rate by traffic channel

Traffic values were cleaned to improve reporting readability:

- **(none)** was converted to **Direct**
- **(not set)** was converted to **Unknown**

5.3 Product Performance & Revenue Drivers

Product-level SQL analysis was performed using nested ecommerce fields to evaluate volume and raw profitability. The analysis included:

- Product validation using UNNEST to unpack nested structural records.
- Top products by absolute purchase count.
- Top products by total revenue generation.

Revenue was calculated by aggregating product revenue fields and converting values into readable currency metrics.

5.4 Conversion Funnel & Drop-Off Analysis

The funnel analysis was performed to track how users move through linear milestones and isolate drop-off friction points:

Stage	Meaning
Product View	User looked at product details
Add to Cart	User showed explicit purchase intent
Purchase	User completed transaction successfully

5.5 Geographic Revenue Distribution

SQL queries were used to isolate regional performance and identify high-value geographic demographics. Analysis included:

- Revenue concentration aggregated by city

- Top revenue-generating physical locations

6. Data Modelling and BigQuery Views

After completing SQL exploration, reusable BigQuery views were created for scalable dashboard reporting. The project contains four analytical views:

1. ecommerce_sales_view
2. traffic_performance_view
3. funnel_analysis_view
4. funnel_chart_view

7. Dashboard Overview

The transformed BigQuery views were connected to Looker Studio to create two interactive dashboard pages for deep-dive reporting:

Page 1: Google Analytics Ecommerce Performance Dashboard

- Revenue trend analysis & Revenue KPI
- Purchase KPI
- Sessions & Conversion rate by traffic channel
- Top products by revenue & Revenue by city

Page 2: Ecommerce Conversion Funnel Analysis

- Product views, Add to cart actions, and Purchases
- Purchase conversion rate
- Daily funnel performance & Ecommerce event trends
- Conversion rate trends

8. Key Business Findings

1. Direct traffic generated the highest number of sessions and strong conversion performance.
2. Organic traffic contributed significant visitors but had lower conversion efficiency.
3. A small number of products generated a large proportion of revenue.
4. Lower-cost branded accessories dominate purchase volume, suggesting customers prefer impulse-friendly merchandise over premium products.
5. While lower priced accessories like stickers dominated purchase frequency, apparel and premium drinkware generated the highest revenue, indicating stronger monetization from higher-priced merchandise categories.

6. The funnel shows a major drop-off between product views and cart additions, with only 33.5% of viewers demonstrating purchase intent. Additionally, cart abandonment is high (~80%), indicating potential friction in the checkout process or price sensitivity.
7. Revenue performance changed over time, showing periods of stronger ecommerce activity.

9. Business Recommendations

- **Optimise high-performing acquisition channels:** Invest further in channels that combine high traffic volume with strong conversion rates.
- **Improve organic conversion:** Analyse landing pages and product pages receiving organic traffic to identify opportunities to improve purchase completion.
- **Focus on high-revenue products:** Promote high-revenue products through targeted campaigns and merchandising strategies.
- **Reduce funnel drop-off:** Investigate reasons for customer drop-off after adding products to carts. Focus engineering and UX improvements on:
 - Checkout complexity and page latency.
 - Hidden shipping costs or rigid delivery timelines.
 - Alternative modern payment gateways.
- **Monitor performance trends:** Continue tracking revenue and conversion trends to identify seasonal patterns and changes in customer behaviour.

10. Conclusion

The project demonstrates practical experience in SQL analytics, cloud data warehousing, dashboard development, and translating data findings into business recommendations. The analysis successfully transforms raw analytics data into actionable insights covering revenue, acquisition channels, products and conversion behaviour.